ED408 — Statistical Machine Learning II Lab

Metaheuristic Optimisation: Simulated Annealing and Genetic Algorithm — Complete Worked Solutions

Problem 1 — Simulated Annealing for Minimisation

Problem. Minimise $f(x) = x^2 - 6x + 13$ using Simulated Annealing with: initial solution $x = 1$, initial temperature $T = 10$, cooling factor $\alpha = 0.5$, minimum temperature $Tmin = 1$, neighbour generation $x' = x \pm 1$ (we move to $x' = x + 1$, the unexplored direction, at every step), and acceptance probability $P = 1$ if $\Delta f \leq 0$, else $P = e^{\wedge}(-\Delta f/T)$, where $\Delta f = f(x') - f(x)$. Random numbers for accepting/rejecting worse solutions, used in order and only when a worse solution appears: $r_1 = 0.30$, $r_2 = 0.80$, $r_3 = 0.20$, $r_4 = 0.90$.

Task 1 — f(x) at the initial solution

$f(1) = 1 - 6 + 13 = 8.$

Task 2 — first neighbour and Δf

$x' = 2$: $f(2) = 4 - 12 + 13 = 5$; $\Delta f = f(2) - f(1) = 5 - 8 = -3.$

Task 3 — is x′ = 2 accepted?

Since $\Delta f = -3 \leq 0$, the acceptance probability is $P = 1$ and the move is **accepted** unconditionally. Because this is an improving (downhill) move, **no random number is consumed**.

Tasks 4 & 5 — four SA iterations and the complete iteration table

The temperature is cooled by $T \leftarrow \alpha T$ after every iteration: $T = 10, 5, 2.5, 1.25$. A random number is drawn (in the given order) only in the rows where $\Delta f > 0$.

Iter	T	Current x	Neighbour x′	f(x)	f(x′)	Δf	P	r	Accept / Reject	Best x (f)
1	10	1	2	8	5	−3	1	—	Accept	2 (5)
2	5	2	3	5	4	−1	1	—	Accept	3 (4)
3	2.5	3	4	4	5	+1	e^(−1/2.5) = 0.6703	r_1 = 0.30	0.30 ≤ 0.6703 → Accept	3 (4)
4	1.25	4	5	5	8	+3	e^(−3/1.25) = 0.0907	r_2 = 0.80	0.80 > 0.0907 → Reject	3 (4)

Iterations 1–2 are downhill and accepted with $P = 1$. Iteration 3 is the first worse move: at $T = 2.5$, $P = e^{\wedge}(-1/2.5) = 0.6703$ and $r_1 = 0.30 \leq P$, so SA **accepts the worse solution x = 4** — the uphill escape that distinguishes SA from greedy descent. Iteration 4: at $T = 1.25$, $P = e^{\wedge}(-3/1.25) = 0.0907$ and $r_2 = 0.80 > P$, so the move is rejected and the algorithm stays at $x = 4$. After iteration 4 the temperature cools to $T = 0.625 < Tmin = 1$ and the algorithm terminates. Random numbers r_3, r_4 are never consumed, because only two worse moves were generated.

Task 6 — best value of x obtained

Tracking the best-so-far solution across the run: **x* = 3, f(x*) = 4** (reached at iteration 2 and never improved upon).

Task 7 — analytical verification

$f'(x) = 2x - 6 = 0 \Rightarrow x = 3$; $f''(x) = 2 > 0$ confirms a minimum, and $f(3) = 9 - 18 + 13 = 4$. The SA result coincides exactly with the analytical minimum, so the algorithm converged correctly.

Problem 2 — Genetic Algorithm Q1: maximise f(x) = x², 0 ≤ x ≤ 31

Encoding. With 5 bits, $2^5 = 32$ strings cover $00000 \to 0$ to $11111 \to 31$ exactly, so the decoded integer is used directly as x. Fitness = $f(x) = x^2$. One full GA generation is executed: fitness → selection probability $f_i/\Sigma f$ → expected count $f_i/\bar{f}$ → actual count (rounded, total fixed at 4) → mating pool → single-point crossover → new fitness.

(a) Exercise population (01100, 11001, 00101, 10011)

S.No	String	x	f(x) = x²	Prob $f_i/\Sigma f$	Expected $f_i/\bar{f}$	Actual count
1	01100	12	144	0.1247	0.499	1
2	11001	25	625	0.5411	2.165	2
3	00101	5	25	0.0216	0.087	0
4	10011	19	361	0.3126	1.250	1

$\Sigma f = 1155, \quad \text{maximum} = 625, \quad \text{average} = 288.75$. The weakest string (00101) receives 0 copies and is eliminated; the strongest (11001) receives 2. **Mating pool:** (01100, 11001, 11001, 10011).

Crossover. Pair (01100 × 11001) at point 4: $0110|0 \times 1100|1 \to 01101, 11000$. Pair (11001 × 10011) at point 2: $11|001 \times 10|011 \to 11011, 10001$.

Offspring	x	fitness x²
01101	13	169
11000	24	576
11011	27	729
10001	17	289

New $\Sigma f = 1763$, new maximum 729, new average 440.75 — one generation lifts the maximum 625 → 729 and the average 288.75 → 440.75. (The notes list 24² as 526; the correct value is 576 — every other entry matches.)

(b) Lecture population (11011, 10001, 01111, 10111) — same machinery

S.No	String	x	f(x) = x²	Prob $f_i/\Sigma f$	Expected $f_i/\bar{f}$	Actual count
1	11011	27	729	0.4114	1.646	2
2	10001	17	289	0.1631	0.652	1
3	01111	15	225	0.1270	0.508	0
4	10111	23	529	0.2985	1.194	1

$\Sigma f = 1772, \quad \text{maximum} = 729, \quad \text{average} = 443$. **Mating pool:** (11011, 11011, 10001, 10111).

Crossover (both pairs at point 2): $11|011 \times 10|111 \to 11111, 10011$ and $11|011 \times 10|001 \to 11001, 10011$.

Offspring	x	fitness x²
11111	31	961
10011	19	361
11001	25	625
10011	19	361

New $\Sigma f = 2308$, new maximum 961, new average 577. The global optimum string 11111 $(x = 31, f = 961)$ already appears in generation 2.

(c) Mutation (bit flipping)

As in the notes: if a string's fitness is very low, flip a bit — e.g. 00011 $(x = 3, f = 9) \to 10011$ $(x = 19, f = 361)$. Mutation's role is to restore lost diversity: once every string starts with the same bits, crossover alone can never reach parts of the search space; a low-probability bit flip re-injects them.

Problem 3 — Genetic Algorithm Q2: maximise f(x) = −x² + 2x, 0 ≤ x ≤ 2

Problem. Population (11010, 00111, 10110, 00101); random numbers 0.4, 0.15, 0.7, 0.9 for roulette-wheel selection; crossover points 1 and 4 (before the 5th digit). **Encoding rule:** $x = \min + (\max - \min)/(2^L - 1) \times (\text{decimal}) = 0 + (2/31)\cdot d$, e.g. $10110 \to d = 22 \to x = 44/31 = 1.419$ (as computed in the notes).

Decoding, fitness, selection probabilities

S.No	String	decimal d	x = 2d/31	f(x) = −x²+2x	prob $f_i/\Sigma f$	cumulative
1	11010	26	1.6774	0.5411	0.2077	0.2077
2	00111	7	0.4516	0.6993	0.2684	0.4760
3	10110	22	1.4194	0.8241	0.3163	0.7923
4	00101	5	0.3226	0.5411	0.2077	1.0000

$\Sigma f = 2.6056$. Because $f(x) = x(2 - x)$ peaks at $x = 1$, the middle strings are fittest: string 3 (x = 1.4194, f = 0.8241) leads.

Roulette-wheel selection with the given random numbers

random r	falls in cumulative interval	selected string
0.40	(0.2077, 0.4760]	2 → 00111
0.15	(0, 0.2077]	1 → 11010
0.70	(0.4760, 0.7923]	3 → 10110
0.90	(0.7923, 1.0000]	4 → 00101

Mating pool: (00111, 11010, 10110, 00101) — every string survives here, but ordered by the wheel.

Crossover at points 1 and 4

Pair (00111 × 11010) at point 1: $0|0111 \times 1|1010 \to 01010, 10111$. Pair (10110 × 00101) at point 4: $1011|0 \times 0010|1 \to 10111, 00100$.

Offspring	decimal	x	f(x)
01010	10	0.6452	0.8741
10111	23	1.4839	0.7659
10111	23	1.4839	0.7659
00100	4	0.2581	0.4495

New $\Sigma f = 2.8554$ (up from 2.6056) and the new best, $x = 0.6452$, $f = 0.8741$, moves toward the analytic optimum $f'(x) = -2x + 2 = 0 \Rightarrow x^* = 1$, $f(x^*) = 1$. This problem also shows why the encoding matters: the 5-bit grid has resolution $2/31 \approx 0.0645$, so the GA can never land closer to the optimum than the grid allows.

Summary

- SA: best $x^* = 3$, $f^* = 4$ after 4 iterations — equals the analytic minimum; the uphill acceptance at iteration 3 (P = 0.6703 vs $r_1 = 0.30$) is the algorithm's defining step.
- GA Q1: one generation improves max fitness 625 → 729 (exercise) and 729 → 961 (lecture, reaching the optimum 11111); selection eliminates the weakest string, crossover recombines the survivors.
- GA Q2: real-interval encoding x = 2d/31 turns the same 5-bit machinery into a continuous-interval search; roulette selection with the given randoms and crossover at points 1 and 4 raise Σf from 2.6056 to 2.8554, approaching $x^* = 1$.
- Full executable implementations of all three problems are in the accompanying notebook SA_GA_Solutions.ipynb.